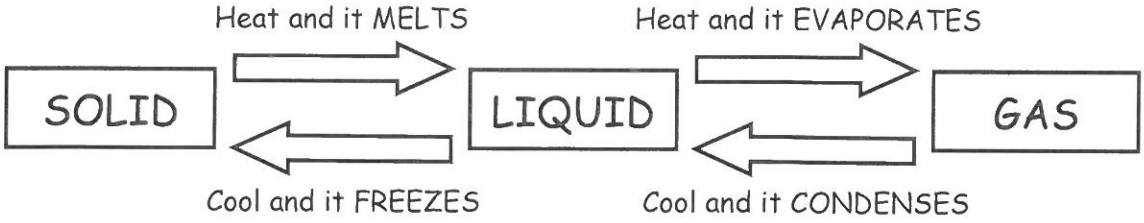
W.S.45. **Changes of state.** Name .....

The three states of matter are SOLID, LIQUID and GAS. One state can change into another. The diagram below shows this.



When a solid changes to a liquid, or a liquid changes to a gas, heat is absorbed. This is because the particles that make up the substance need more energy to move faster and overcome the forces that hold them together. When a gas changes to a liquid, or a liquid changes to a solid, heat is given out. This is because the particles lose energy as they slow down. The substance still keeps the SAME MASS because it still contains the SAME NUMBER OF PARTICLES.

Exercise - Use the information in the table below to help you complete the sentences at the bottom of this page.

Substance	Melting point (°C)	Boiling point (°C)
Oxygen	-219	-183
Ethanol	-15	78
Water	0	100
Sulphur	119	445
Iron	1,540	2,900

1) Oxygen is a _ _ _ at room temperature.

2) Water and _ _ _ _ _ _ _ are liquids at room temperature.

3) _ _ _ _ _ _ _ and iron are solids at room temperature.

4) Sulphur melts at a temperature of _____ °C

5) Iron melts at a temperature of _____ °C

6) The substance with the lowest melting point in the table is _ _ _ _ _ _

7) Ethanol has a _ _ _ _ _ boiling point than water.

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